18 The function f is given by

$$f(x) = 2x^3 - 4$$

(a) Show that $f^{-1}(50) = 3$

$f(3) = 2(3^3) - 4 = 54 - 4 = 50.$

So $f^{-1}(50) = 3$, as required.

2 marks total
Evaluate f(3)
Conclude inverse value

(2)

The functions g and h are given by

$$g(x) = x + 2 \text{ and } h(x) = x^2$$

(b) Find the values of x for which

$$hg(x) = 3x^2 + x - 1$$

$hg(x) = h(x + 2) = (x + 2)^2.$

$x^2 + 4x + 4 = 3x^2 + x - 1$

$2x^2 - 3x - 5 = 0$

$(2x - 5)(x + 1) = 0$

$x = 5/2$ or $x = -1.$

4 marks total
Compose h after g
Expand and rearrange
Factor the quadratic
Both roots

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(4)

(Total for Question 18 is 6 marks)

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